

COM Momentum And Collisions

Questions with Answer Keys

JEE Main 2020 Chapterwise

Physics

Q1 JEE Main 2020 - 2 September (Morning)

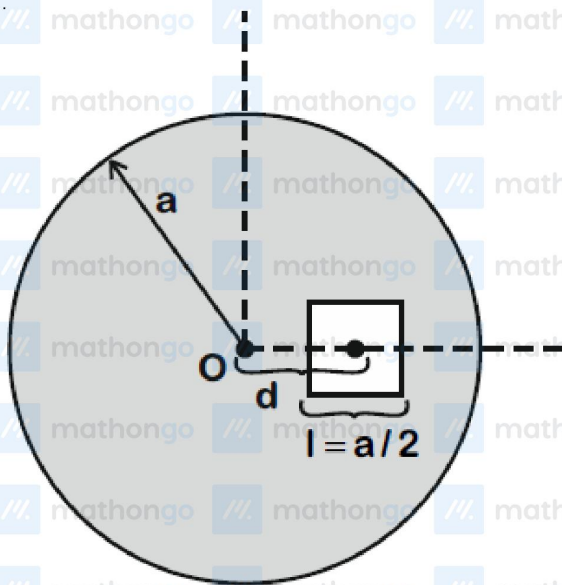
A particle of mass m with an initial velocity u collides perfectly elastically with a mass $3m$ at rest. It moves with a velocity v after collision, then, v is given by

- (A) $v = \frac{1}{\sqrt{6}}u$
- (B) $v = \frac{u}{\sqrt{3}}$
- (C) $v = \sqrt{\frac{2}{3}}u$
- (D) $v = \frac{u}{\sqrt{2}}$

Q2 JEE Main 2020 - 2 September (Evening)

A square shaped hole of side $l = \frac{a}{2}$ is carved

out at a distance $d = \frac{a}{2}$ from the centre 'O' of a uniform circular disk of radius a . If the distance of the centre of mass of the remaining portion from O is $-\frac{a}{x}$, value of x (to the nearest integer) is



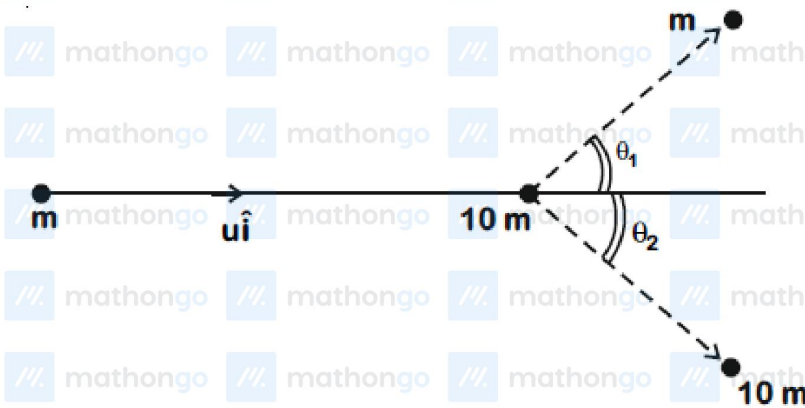
Q3 JEE Main 2020 - 2 September (Evening)

A particle of mass m is moving along the x -axis with initial velocity u . It collides elastically with a particle of mass $10m$ at rest and then moves with half its initial kinetic energy (see figure). If

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$\sin \theta_1 = \sqrt{n} \sin \theta_2$ then value of n is



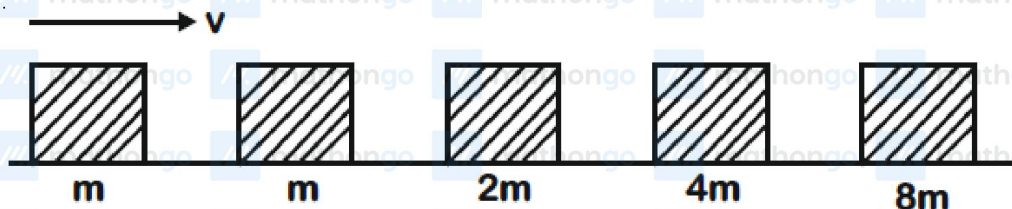
Q4 JEE Main 2020 - 3 September (Evening)

A block of mass 1.9 kg is at rest at the edge of a table, of height 1 m . A bullet of mass 0.1 kg collides with the block and sticks to it. If the velocity of the bullet is 20 m/s in the horizontal direction just before the collision then the kinetic energy just before the combined system strikes the floor, is [Take $g = 10\text{ m/s}^2$. Assume there is no rotational motion and loss of energy after the collision is negligible.]

- (A) 19 J
- (B) 23 J
- (C) 20 J
- (D) 21 J

Q5 JEE Main 2020 - 4 September (Morning)

Blocks of masses $m, 2m, 4m$ and $8m$ are arranged in a line on a frictionless floor. Another block of mass m , moving with speed v along the same line (see figure) collides with mass m in perfectly inelastic manner. All the subsequent collisions are also perfectly inelastic. By the time the last block of mass $8m$ starts moving the total energy loss is $p\%$ of the original energy. Value of ' p ' is close to



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(A) 37

(B) 77

(C) 87

(D) 94

Q6 JEE Main 2020 - 5 September (Evening)

A spaceship in space sweeps stationary interplanetary dust. As a result, its mass increases at a rate

$\frac{dM(t)}{dt} = bv^2(t)$, where $v(t)$ is its instantaneous velocity. The instantaneous acceleration of the

satellite is

(A) $-bv^3(t)$ (B) $-\frac{2bv^3}{M(t)}$ (C) $-\frac{bv^3}{M(t)}$ (D) $-\frac{bv^3}{2M(t)}$ **Q7 JEE Main 2020 - 6 September (Morning)**

Two bodies of the same mass are moving with the same speed, but in different directions in a plane.

They have a completely inelastic collision and move together thereafter with a final speed which is half of their initial speed. The angle between the initial velocities of the two bodies (in degree) is

Q8 JEE Main 2020 - 6 September (Evening)

Particle A of mass m_1 moving with velocity $(\sqrt{3}\hat{i} + \hat{j})ms^{-1}$ collides with another particle B of mass m_2 which is at rest initially. Let $\vec{v}_1$ and $\vec{v}_2$ be the velocities of particles A and B after collision

respectively. If $m_1 = 2m_2$ and after collision $\vec{v}_1 = (\hat{i} + \sqrt{3}\hat{j})ms^{-1}$, the angle between $\vec{v}_1$ and $\vec{v}_2$ is

(A) -45° (B) 60° (C) 15° (D) 105°

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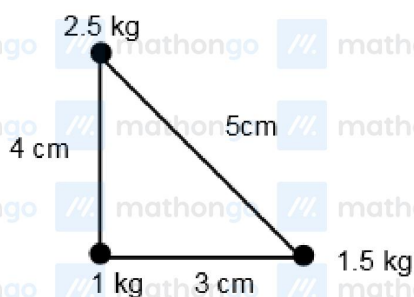
Physics

Q9 JEE Main 2020 - 6 September (Evening)

The centre of mass of a solid hemisphere of radius 8 cm is $x\text{ cm}$ from the centre of the flat surface. Then value of x is

Q10 JEE Main 2020 - 7 January (Morning)

Three point particles of masses 1.0 kg , 1.5 kg and 2.5 kg are placed at three corners of a right angle triangle of sides 4.0 cm , 3.0 cm and 5.0 cm as shown in the figure. The center of mass of the system is



at a point :

- (A) 0.6 cm to the right and 2 cm above 1 kg mass
- (B) 0.9 cm to the right and 2 cm above 1 kg mass
- (C) 0.9 cm to the left and 2 cm above 1 kg mass
- (D) 0.9 cm to the right and 1.5 cm above 1 kg mass

Q11 JEE Main 2020 - 7 January (Evening)

Consider uniform cubical box of side a on a rough floor that is to be moved by applying minimum possible force F at a point b above its centre of mass (see figure). If the coefficient of friction is

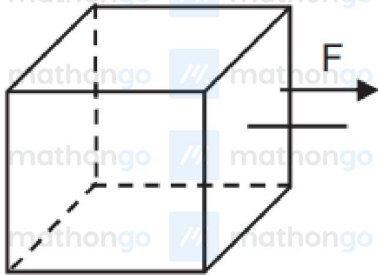
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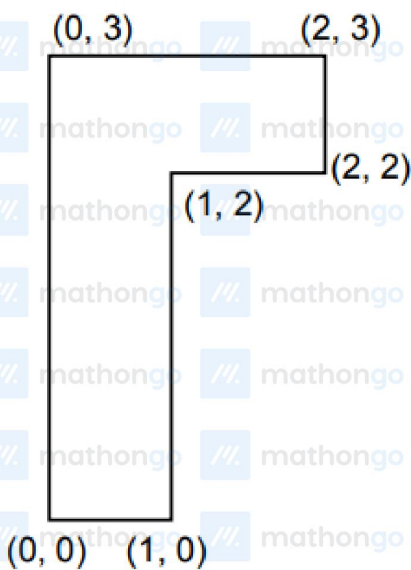
Physics

$\mu = 0.4$, the maximum possible value of $100 \times \frac{b}{a}$ for box not to topple before moving is



Q12 JEE Main 2020 - 8 January (Morning)

Find the coordinates of centre of mass of the lamina, shown in figure.



- (A) 0.75, 1.75
- (B) 0.75, 1.5
- (C) 0.5, 1.75
- (D) 0.5, 1.5

Q13 JEE Main 2020 - 8 January (Morning)

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Physics

A body A , of mass $m = 0.1\text{ kg}$ has an initial velocity of $3\hat{i}\text{ ms}^{-1}$. It collides elastically with another body, B of the same mass which has an initial velocity of $5\hat{j}\text{ ms}^{-1}$. After collision, A moves with a velocity $\vec{v} = 4(\hat{i} + \hat{j})$. The energy of B after collision is written as $\frac{x}{10}\text{ J}$. The value of x is

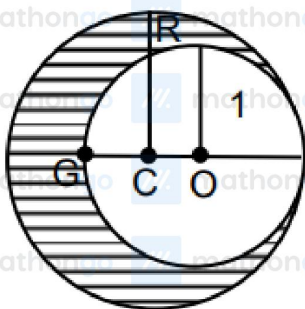
Q14 JEE Main 2020 - 8 January (Evening)

A particle of mass m is dropped from a height h above the ground. At the same time another particle of the same mass is thrown vertically upwards from the ground with a speed of $\sqrt{2gh}$. If they collide head-on completely in elastically, the time taken for the combined mass to reach the ground, in units of $\sqrt{\frac{h}{g}}$ is

- (A) $\sqrt{\frac{3}{2}}$
- (B) $\sqrt{\frac{1}{2}}$
- (C) $\sqrt{3}$
- (D) $\sqrt{\frac{1}{5}}$

Q15 JEE Main 2020 - 8 January (Evening)

As shown in figure. When a spherical cavity (centred at O) of radius 1 is cut out of a uniform sphere of radius R (centred at C), the centre of mass of remaining (shaded) part of sphere is at G , i.e on the surface of the cavity. R can be determined by the equation :



- (A) $(R^2 + R + 1)(2 - R) = 1$
- (B) $(R^2 - R - 1)(2 - R) = 1$

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(C) $(R^2 - R + 1)(2 - R) = 1$

(D) $(R^2 + R - 1)(2 - R) = 1$

Q16 JEE Main 2020 - 9 January (Morning)

Two particles of equal mass m have respective initial velocities $u\hat{i}$ and $u\left(\frac{i+j}{2}\right)$. They collide completely inelastically. The energy lost in the process is :

(A) $\frac{mv^2}{8}$

(B) $\frac{5mv^2}{8}$

(C) $\frac{mv^2}{4}$

(D) $\frac{3vm^2}{8}$

Q17 JEE Main 2020 - 9 January (Evening)

A rod of length L has non-uniform linear mass density given by $\rho(x) = a + b\left(\frac{x}{L}\right)^2$, where a and b are constants and $0 \leq x \leq L$. The value of x for the centre of mass of the rod is at :

(A) $\frac{4}{3} \left(\frac{a+b}{2a+3b} \right) L$

(B) $\frac{3}{4} \left(\frac{2a+b}{3a+b} \right) L$

(C) $\frac{3}{2} \left(\frac{2a+b}{3a+b} \right) L$

(D) $\frac{3}{2} \left(\frac{a+b}{2a+b} \right) L$

Q18 JEE Main 2020 - 9 January (Evening)

A particle of mass m is projected with a speed u from the ground at an angle $\theta = \pi/3$ w.r.t. horizontal (x -axis). When it has reached its maximum height, it collides completely in-elastically with another particle of the same mass and velocity $u\hat{i}$. The horizontal distance covered by the combined mass before reaching the ground is :

(A) $\frac{3\sqrt{3}}{8} \frac{u^2}{g}$

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(B) $2\sqrt{2}\frac{u^2}{g}$

(C) $\frac{5}{8}\frac{u^2}{g}$

(D) $\frac{3\sqrt{2}}{4}\frac{u^2}{g}$

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Physics

Answer Key

Q1 (D)

Q2 (23)

Q3 (10)

Q4 (D)

Q5 (D)

Q6 (C)

Q7 (120)

Q8 (D)

Q9 (3)

Q10 (B)

Q11 (50)

Q12 (A)

Q13 (1)

Q14 (A)

Q15 (A)

Q16 (A)

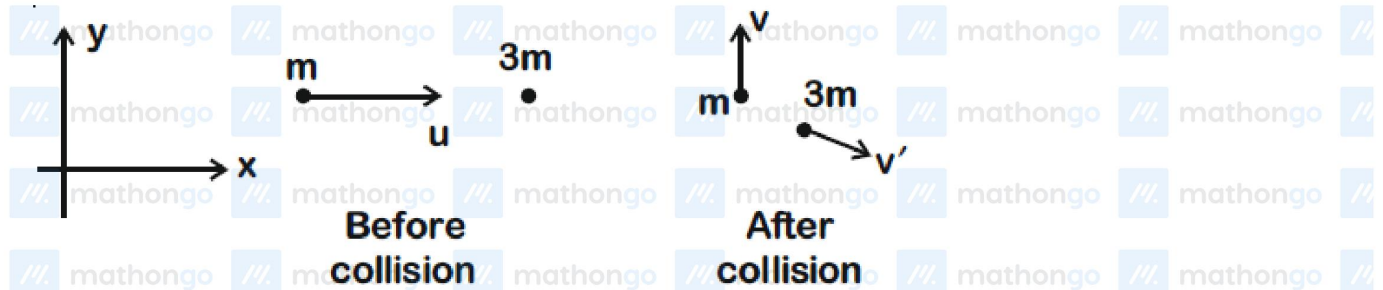
Q17 (B)

Q18 (A)

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Q1



Conservation of linear momentum

$$3mv'' = mu_i - mv_j$$

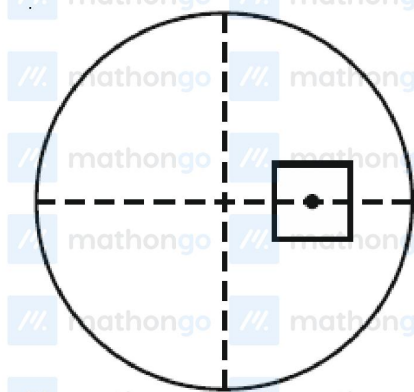
for elastic collision

$$\frac{1}{2}3mv'^2 + \frac{1}{2}mv^2 = \frac{1}{2}mu^2$$

Eliminating v' gives

$$v = \frac{u}{\sqrt{2}}$$

Q2

Let the density is σ Then original mass $m_0 = \pi a^2 \sigma$

$$\text{Remaining mass } m' = \left(\pi a^2 - \frac{a^2}{4} \right) \sigma = a^2 \left(\frac{4\pi - 1}{4} \right) \sigma$$

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Hints & Solutions

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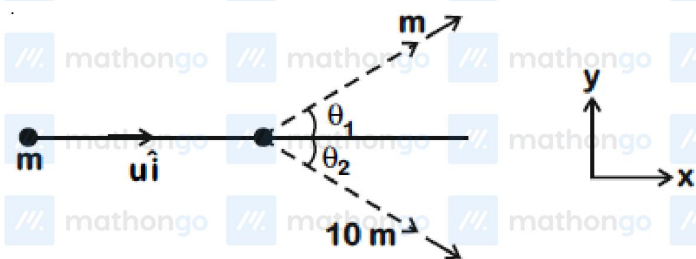
Physics

$$\text{Removed mass } m = \frac{a^2}{4} \cdot \sigma$$

$$\therefore \pi a^2 \sigma(0) = \frac{a^2}{4} \cdot \sigma \times \frac{a}{2} + \frac{\sigma a^2}{4} (4\pi - 1)r$$

$$\Rightarrow r = \frac{-a}{2(4\pi - 1)} = \frac{-a}{23.13} \approx \frac{-a}{23}$$

Q3



since energy of m reduced by half $\therefore u_1 = \frac{u}{\sqrt{2}}$

$$\text{and } \frac{1}{2} \times 10m \cdot v_1^2 = \frac{1}{2} m \frac{u^2}{2} \Rightarrow v_1 = \frac{u}{\sqrt{20}}$$

Now, momentum in y direction will remain conserved

$$\therefore mu_1 \sin \theta_1 = 10mv_1 \sin \theta_2$$

$$\Rightarrow \frac{u}{\sqrt{2}} \sin \theta_1 = 10 \frac{u}{\sqrt{20}} \sin \theta_2$$

$$\Rightarrow \sin \theta_1 = \sqrt{10} \sin \theta_2$$

Q4

Velocity of block after collision

$$= \frac{0.1 \times 20}{1.9 + 0.1} = 1 \text{ m/s}$$

Kinetic energy just before striking the floor

$$= \frac{1}{2} \times 2 \times 1^2 + 2 \times 10 \times 1$$

$$= 21 \text{ J}$$

Q5

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Physics

$$\vec{P}_i = \vec{P}_f$$

$$\Rightarrow mv = (16m)v_0 \Rightarrow v_0 = \frac{v}{16}$$

$$(\Delta KE)_{\text{Loss}} = \frac{1}{2}mv^2 - \frac{1}{2}(16m)\left(\frac{v}{16}\right)^2$$

$$\Rightarrow \% \text{ Loss} = \frac{(\Delta KE)_{\text{Loss}}}{\frac{1}{2}mv^2} \times 100 = \left(1 - \frac{1}{16}\right) \times 100$$

$$\simeq 94\%$$

Q6

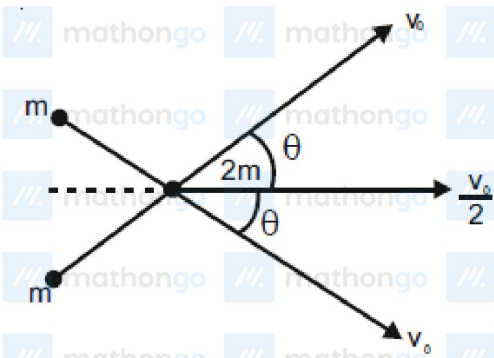
$$F = \frac{dP}{dt} = \left(\frac{dM}{dt}\right)v$$

$$\Rightarrow M(t)a = -(bv^2)v$$

$$\Rightarrow a = -\frac{bv^3}{M(t)}$$

Q7

$$mv_0 \times \cos \theta \times 2 = 2m \times \left(\frac{v_0}{2}\right)$$



$$\Rightarrow \cos \theta = \frac{1}{2}$$

$$\Rightarrow \theta = 60^\circ$$

$$\therefore 2\theta = 120^\circ$$

Q8

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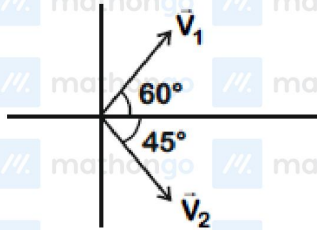
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Physics



$$m_2 = m \text{ and } m_1 = 2m$$

$$\vec{p}_i = \vec{p}_f$$

$$\Rightarrow 2m(\sqrt{3}\hat{i} + \hat{j}) = 2m(\hat{i} + \sqrt{3}\hat{j}) + m\vec{v}_2$$

$$\Rightarrow \vec{v}_2 = (2\sqrt{3} - 2)\hat{i} - \hat{j}(2\sqrt{3} - 2)$$

$$\vec{v}_1 = (\hat{i} + \sqrt{3}\hat{j})$$

$$\Rightarrow \text{Angle } \theta = 105^\circ$$

Q9

$$x = \frac{3R}{8} = 3\text{cm}$$

Q10

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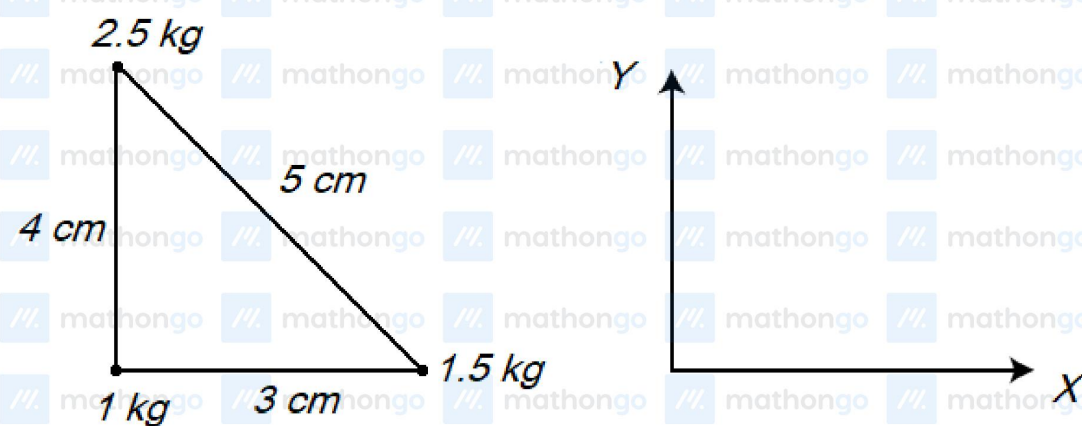
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Physics

Take 1 kg mass at origin

$$X_{cm} = \frac{1 \times 0 + 1.5 \times 3 + 2.5 \times 0}{5} = 0.9 \text{ cm}$$

$$Y_{cm} = \frac{1 \times 0 + 1.5 \times 0 + 2.5 \times 4}{5} = 2 \text{ cm}$$



Q11

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Physics

For no toppling

$$F \left(\frac{a}{2} + b \right) \leq mg \frac{a}{2}$$

$$\mu \frac{a}{2} + \mu b \leq \frac{a}{2}$$

$$0.2a + 0.4b \leq 0.5a$$

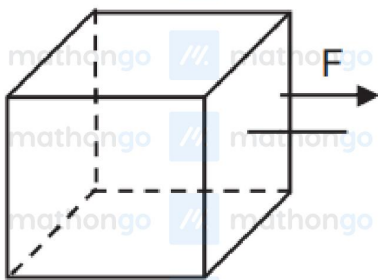
$$0.4b \leq 0.3a$$

$$b \leq \frac{3a}{4}$$

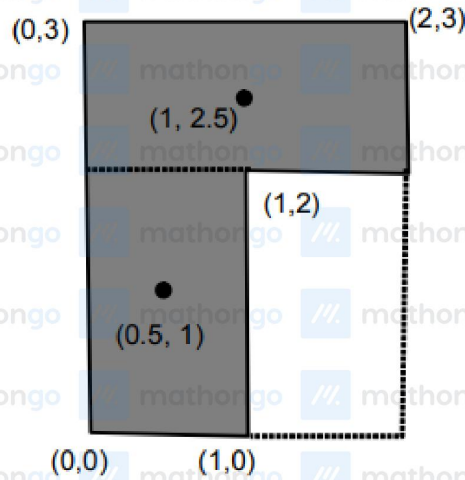
$$b \leq 0.75a \quad (\text{in limiting case})$$

But it is not possible as b can maximum be equal to $0.5a$ only

$$\therefore \left(100 \frac{b}{a} \right)_{\max} = 50.00$$



Q12



$$\vec{r}_{cm} = \frac{1 \times \left(\frac{\hat{i}}{2} + \hat{j} \right) + 1 \times \left(\hat{i} + \frac{5\hat{j}}{2} \right)}{2}$$

$$\vec{r}_{cm} = \frac{3}{4}\hat{i} + \frac{7}{4}\hat{j}$$

Q13

For elastic collision $KE_i = KE_f$

$$\frac{1}{2}m \times 25 + \frac{1}{2} \times m \times 9 = \frac{1}{2}m \times 32 + \frac{1}{2}mv^2$$

$$34 = 32 + v^2$$

$$KE = \frac{1}{2} \times 0.1 \times 2 = 0.1 \text{ J} = \frac{1}{10}$$

$$x = 1$$

Q14

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Physics

time for collision $t_1 = \frac{g}{\sqrt{2gh}}$

After t_1

$$V_A = 0 - gt_1 = -\sqrt{\frac{gh}{2}}$$

$$\text{and } V_B = \sqrt{2gh} - gt_1 = \sqrt{gh} \left[\sqrt{2} - \frac{1}{\sqrt{2}} \right]$$

at the time of collision

$$\vec{P}_i = \vec{P}_f$$

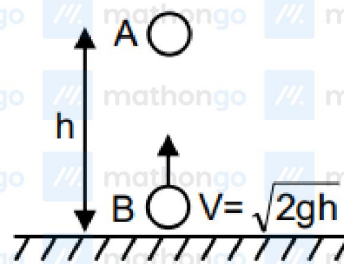
$$\Rightarrow m\vec{V}_A + m\vec{V}_B = 2m\vec{V}_f$$

$$\Rightarrow -\sqrt{\frac{gh}{2}} + \sqrt{gh} \left[\sqrt{2} - \frac{1}{\sqrt{2}} \right] = 2\vec{V}_f$$

$$V_f = 0$$

$$\text{and height from ground} = h - \frac{1}{2}gt_1^2 = h - \frac{h}{4} = \frac{3h}{4}$$

$$\text{so time} = \sqrt{2 \times \frac{\left(\frac{3h}{4}\right)}{g}} = \sqrt{\frac{3h}{2g}}$$



Q15

$$M_1 = \frac{4}{3}\pi R^3 \rho$$

$$M_2 = \frac{4}{3}\pi(1)^3(-\rho)$$

$$X_{com} = \frac{M_1 X_1 + M_2 X_2}{M_1 + M_2}$$

$$\Rightarrow \frac{\left[\frac{4}{3}\pi R^3 \rho\right] 0 + \left[\frac{4}{3}\pi(1)^3(-\rho)\right] [R-1]}{\frac{4}{3}\pi R^3 \rho + \frac{4}{3}\pi(1)^3(-\rho)} = -(2-R)$$

$$\Rightarrow \frac{(R-1)}{(R^3-1)} = (2-R) \quad (R \neq 1)$$

$$\frac{(R-1)}{(R-1)(R^2+R+1)} = 2-R$$

$$(R^2+R+1)(2-R) = 1$$

Q16

Conserving momentum

$$mv \hat{i} + m\left(\frac{v}{2}\hat{i} + \frac{v}{2}\hat{j}\right) = 2m(v_1 \hat{i} + v_2 \hat{j})$$

on solving

$$v_1 = \frac{3v}{4} \text{ and } v_2 = \frac{v}{4}$$

Change in K.E.

$$\left[\frac{1}{2}mv^2 + \frac{1}{2}m\left(\frac{v}{2}\sqrt{2}\right)^2\right] - \left[\frac{1}{2}(2m)\left(\frac{9v^2}{16} + \frac{v^2}{16}\right)\right]$$

$$= \frac{3mv^2}{4} - \frac{5mv^2}{8} = \frac{mv^2}{8}$$

Q17

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Physics

$$x_{cm} = \frac{1}{M} \int_0^{\ell} x \cdot dM$$

$$dM = \lambda \cdot dx = \left(a + b \left(\frac{x}{\ell} \right)^2 \right) \cdot dx$$

$$x_{cm} = \frac{\int x dM}{\int dM} = \frac{\int x \lambda dx}{\int \lambda dx} = \frac{\int_0^{\ell} x \left(a + \frac{bx^2}{\ell^2} \right) dx}{\int_0^{\ell} \left(a + \frac{bx^2}{\ell^2} \right) dx}$$

$$= \frac{a \left(\frac{x^2}{2} \right)_0^{\ell} + \frac{b}{\ell^2} \left(\frac{x^4}{4} \right)_0^{\ell}}{a(x)_0^{\ell} + \frac{b}{\ell^2} \left(\frac{x^3}{3} \right)_0^{\ell}}$$

$$= \frac{a \frac{\ell^2}{2} + b \frac{\ell^2}{4}}{a\ell + \frac{b\ell}{3}} = \frac{(2a+b)\ell}{(3a+b)4} \times 3$$

$$= \frac{3\ell}{4} \left(\frac{2a+b}{3a+b} \right)$$

$$= \frac{3\ell}{4} \left(\frac{2a+b}{3a+b} \right)$$



Q18

$$mu + mu \cos \theta = 2mv$$

$$\Rightarrow v = \frac{u(1 + \cos 60^\circ)}{2} = \frac{3}{4}u$$

so horizontal range after collision = vt

$$= v \sqrt{\frac{2H_{\max}}{g}}$$

$$= \frac{3}{4}u \sqrt{\frac{2u^2 \sin^2 (60^\circ)}{2g^2}}$$

$$= \frac{3}{4}u^2 \frac{\sqrt{\frac{3}{4}}}{g} = \frac{3\sqrt{3}u^2}{8g}$$

